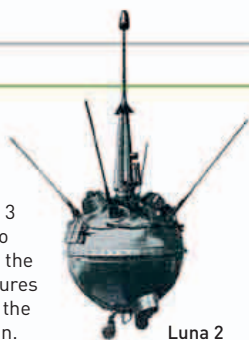


1957
Man-made satellite
On October 4, USSR launches the first satellite into orbit around Earth, Sputnik 1. Today there are over 500 working satellites.

1959
Luna 2 and 3
The Soviet lunar probes Luna 2 and 3 are the first craft to successfully reach the Moon. Luna 3 captures the first images of the far side of the Moon.



Luna 2

1962
Mission to Venus
On December 14, the Mariner 2 becomes the first spacecraft to fly past another planet, revealing Venus as a hothouse planet.

1963
First female cosmonaut
On June 7, Soviet cosmonaut Valentina Tereshkova becomes the first woman in space. A crater on the far side of the Moon is named after her. The first US female astronaut was Sally Ride, 20 years later.



Tereshkova

1966
Landing on the Moon
On February 3, Soviet probe Luna 9 becomes the first spacecraft to successfully land on the Moon. On May 30, the US Surveyor 1, made the second soft landing.

1949
Animals in space
The first astronauts were animals. American rocket scientists send Rhesus monkey Albert II into space in 1949. Soviet dog Laika became the first animal to orbit Earth in 1957.



Laika

1961
First person in space
On April 12, Soviet cosmonaut Yuri Gagarin becomes the first person in space. He completes one orbit of the Earth in Vostok 1. In May, Alan Shepard (1923–98) becomes the first American in space.



Yuri Gargarin

1965
First spacewalks
On March 18, Soviet Aleksei Leonov becomes the first astronaut to venture outside his craft. American astronaut Edward White completes a spacewalk in June.



Edward White

THE STORY OF SPACE EXPLORATION

FROM THE FIRST ARTIFICIAL SATELLITE TO REACHING OTHER WORLDS AND THE EDGE OF THE SOLAR SYSTEM

The launch of the Soviet Union's Sputnik 1 in October 1957 is usually taken to mark the start of space exploration, even though some earlier flights had left Earth's atmosphere. It was the beginning of a series of adventures that has taken astronauts to the Moon and probes to distant planets.

Sputnik 1 was followed a month later by the voyage of the Soviet dog Laika, which became the first animal to orbit Earth. The first human to travel in space was Yuri Gagarin, who orbited Earth in the Soviet spacecraft Vostok 1 in April 1961. These Soviet successes threw down the challenge to the US to step up its space exploration programme. In 1965, the American Mariner 4 sent back the first close-up pictures of another planet, Mars. In 1966, a Soviet probe, Luna 9, made a soft-landing on the Moon and sent back the first pictures from the surface. Three years later, American

astronauts Neil Armstrong and Buzz Aldrin set foot on the Moon. Their first moments there were broadcast live to people around the world. Astronauts returned to the Moon several times in the 1970s, but most voyages of exploration since have been by robot craft, which have now traveled to every planet in the Solar System and some even beyond that (see panel, right).



hatch through which astronauts entered and exited module

Apollo 11 command module
This was part of the spacecraft that carried Neil Armstrong, Buzz Aldrin, and Michael Collins on their historic mission to the Moon in 1969.



VOYAGER 1 AND 2

When the two Voyager probes were launched in 1977, they were expected to send back useful data from as far as Jupiter and Saturn. However, they have continued to return data as they traveled through the heliosheath—the very edge of the Solar System—and will soon be in interstellar space. Voyager 1 may well have left the Solar System in October 2012. It is the most distant human-made object, nearly 11,495 billion miles (18.5 billion km) away in March 2013. Voyager 2 is not far behind, over 9,320 billion miles (15 billion km) away.

“ IF OUR LONG-TERM **SURVIVAL IS AT STAKE**, WE HAVE A BASIC RESPONSIBILITY TO OUR SPECIES TO **VENTURE TO OTHER WORLDS.** ”

Carl Sagan, American cosmologist, 1934–96

Apollo 11 hatch

The main hatch on the Apollo Command Module had to provide a perfect seal to protect the crew. It was redesigned to open outward after an accident in 1967 in which astronauts were trapped inside the capsule as it caught fire.